

Flipkart Analytics Dashboard

Python Data Cleaning - SQL Server Analysis - Power BI Dashboard

Domain Background

E-commerce platforms generate large volumes of structured product data such as product names, brands, categories, prices, discounts, ratings, images, and product specifications. This data helps businesses understand product mix, pricing strategies, customer value, premium segments, and discount effectiveness.

Why This Data Is Important

- It helps identify high-value and premium products.
- It supports category-wise and brand-wise product analysis.
- It helps understand discount behavior and customer savings.
- It supports pricing strategy and outlier detection.
- It helps build business dashboards for decision making.

About the Dataset

Type	Retail / E-commerce Product Dataset
Nature	Structured tabular data
Source	Kaggle Flipkart product sample dataset
Final cleaned records	20,001 products
Final columns	17 columns
Tools used	Python Pandas, SQL Server, Power BI

Business Problem

How can Flipkart product data be analyzed to understand product pricing, discounts, categories, brands, ratings, and premium outlier products in order to support better e-commerce business decisions?

Breaking Down the Problem

Product Pricing Understanding

- Which products are expensive?
- Which products are cheaper?
- Which products are above the category average price?

Discount Strategy Analysis

- Which products offer the highest discount?
- Which categories provide higher average discounts?
- Which products provide maximum savings?

Brand and Category Analysis

- Which brands have the most products?
- Which categories dominate product count?
- Which brands show strong rating and discount behavior?

Product Rating Analysis

- How are products distributed by rating?
- Which products are high rated?
- How can unrated products be handled professionally?

Outlier Product Detection

- Which products have unusually high prices?
 - How many price outliers exist?
 - Which premium products need separate analysis?
-

Data Dictionary

Column Name	Description
uniq_id	Unique product row identifier
crawl_datetime	Date and time when product data was collected
product_url	Product web page URL
product_name	Name of the product listed on Flipkart
product_category_tree	Full category hierarchy of the product
pid	Product ID
retail_price	Original listed product price
discounted_price	Selling price after discount
image	Product image URL
is_fk_advantage_product	Indicates whether Flipkart Advantage is available
description	Product description text
product_rating	Customer product rating
overall_rating	Overall rating of product
brand	Product brand name
product_specifications	Structured product specifications
discount_percentage	Feature engineered discount percentage
price_outlier	Feature engineered flag for price outliers

Project Objectives

- Clean the raw Flipkart dataset using Python and Pandas.
- Handle missing values, duplicates, data types, and text inconsistencies.
- Create business features such as discount percentage and price outlier flag.
- Load cleaned data into SQL Server for business analysis.
- Execute SQL queries to answer product, brand, discount, rating, and pricing questions.
- Build a Power BI dashboard with KPIs, filters, charts, treemap, and sales overview.
- Derive actionable business insights from product analytics.

EDA and Data Cleaning using Python

Python was used to bring raw data into a structured DataFrame, inspect data quality, clean missing values, remove duplicates, standardize columns, create new analytical features, and prepare the dataset for SQL Server and Power BI.

Step	Description
Data Loading	Loaded CSV using pandas read_csv and viewed initial records using head.
Initial Exploration	Checked shape, columns, data types, and missing values.
Missing Values	Filled numeric prices using median and categorical values using unknown/default categories.
Duplicate Handling	Detected and removed one duplicate row.
Data Type Fixing	Converted timestamp and rating fields into usable formats.
Text Standardization	Converted text columns to lowercase and stripped extra spaces.
Feature Engineering	Created discount_percentage and price_outlier columns.
Outlier Detection	Used IQR method to identify 2,063 price outlier products.
Export	Exported final cleaned data and loaded it into SQL Server.

Python Notebook Screenshots

Dataset Import and Initial View

Imported libraries and loaded the Flipkart CSV dataset into a Pandas DataFrame.

```
import pandas as pd
import numpy as np
import re
```

```
df = pd.read_csv("flipkart_com-ecommerce_sample.csv")
df.head()
```

	uniq_id	crawl_timestamp	product_url	product_name	product_category_tree	pid	retail_price
0	c2d766ca982eca8304150849735ffef9	2016-03-25 22:59:23 +0000	http://www.flipkart.com/alisha-solid-women-s-c...	Alisha Solid Women's Cycling Shorts	["Clothing >> Women's Clothing >> Lingerie, SL...	SRTEH2FF9KEDEFGF	999.0
1	7f7036a6d550aaa89d34c77bd39a5e48	2016-03-25 22:59:23 +0000	http://www.flipkart.com/fabhomedecor-fabric-do...	FabHomeDecor Fabric Double Sofa Bed	["Furniture >> Living Room Furniture >> Sofa B...	SBEEH3QGU7MFYJFY	32157.0
2	f449ec65dcbc041b6ae5e6a32717d01b	2016-03-25 22:59:23 +0000	http://www.flipkart.com/aw-bellies/p/itmeh4grg...	AW Bellies	["Footwear >> Women's Footwear >> Ballerinas >...	SHOEH4GRSUBJGZXE	999.0
3	0973b37acd0c664e3de26e97e5571454	2016-03-25 22:59:23 +0000	http://www.flipkart.com/alisha-solid-women-s-c...	Alisha Solid Women's Cycling Shorts	["Clothing >> Women's Clothing >> Lingerie, SL...	SRTEH2F6HUZMQ6SJ	699.0
4	bc940ea42ee6bef5ac7cea3fb5cfbee7	2016-03-25 22:59:23 +0000	http://www.flipkart.com/sicons-all-purpose-arn...	Sicons All Purpose Arnica Dog Shampoo	["Pet Supplies >> Grooming >> Skin & Coat Care...	PSOEH3ZYDMSYARJ5	220.0

First Look: Shape, Columns, Data Types and Missing Values

Checked structure and missing values before cleaning.

```
# Shape of dataset
print(df.shape)

# Column names
print(df.columns)

# Data types
print(df.dtypes)

# Missing values
print(df.isnull().sum())
```

```
(20002, 15)
Index(['uniq_id', 'crawl_timestamp', 'product_url', 'product_name',
       'product_category_tree', 'pid', 'retail_price', 'discounted_price',
       'image', 'is_FK_Advantage_product', 'description', 'product_rating',
       'overall_rating', 'brand', 'product_specifications'],
      dtype='object')
```

uniq_id	object
crawl_timestamp	object
product_url	object
product_name	object
product_category_tree	object
pid	object
retail_price	float64
discounted_price	float64
image	object
is_FK_Advantage_product	object
description	object
product_rating	object
overall_rating	object
brand	object
product_specifications	object
dtype: object	
uniq_id	2
crawl_timestamp	2
product_url	2
product_name	2
product_category_tree	2
pid	2
retail_price	80
discounted_price	80
image	5
is_FK_Advantage_product	2
description	4
product_rating	2
overall_rating	2
brand	5866
product_specifications	16
dtype: int64	

Column Name Standardization

Converted column names into lowercase and SQL-friendly naming format.

```
# Clean column names

df.columns = (
    df.columns
    .str.strip()
    .str.lower()
    .str.replace(" ", "_")
)

# Check updated column names
print(df.columns)

Index(['uniq_id', 'crawl_timestamp', 'product_url', 'product_name',
       'product_category_tree', 'pid', 'retail_price', 'discounted_price',
       'image', 'is_fk_advantage_product', 'description', 'product_rating',
       'overall_rating', 'brand', 'product_specifications'],
      dtype='object')
```

Duplicate Detection

Checked the dataset for repeated rows before removing them.

```
# Check duplicate rows

print(df.duplicated().sum())
```

1

Screenshot: Duplicate Detection

Duplicate Removal

Removed duplicate rows and verified the dataset shape.

```
# Remove duplicate rows

df = df.drop_duplicates()

# Verify again
print(df.duplicated().sum())

# Check new shape
print(df.shape)
```

0
(20001, 15)

Screenshot: Duplicate Removal

Missing Value Percentage Analysis

Calculated missing count and missing percentage for each column.

```
missing_percent = (df.isnull().sum() / len(df)) * 100

missing_summary = pd.DataFrame({
    'missing_count': df.isnull().sum(),
    'missing_percentage': missing_percent
})

print(missing_summary[missing_summary['missing_count'] > 0])
```

	missing_count	missing_percentage
uniq_id	1	0.005000
crawl_timestamp	1	0.005000
product_url	1	0.005000
product_name	1	0.005000
product_category_tree	1	0.005000
pid	1	0.005000
retail_price	79	0.394980
discounted_price	79	0.394980
image	4	0.019999
is_fk_advantage_product	1	0.005000
description	3	0.014999
product_rating	1	0.005000
overall_rating	1	0.005000
brand	5865	29.323534
product_specifications	15	0.074996

Screenshot: Missing Value Percentage Analysis

Missing Value Handling

Filled numeric, text, rating, and brand missing values based on business logic.

```
[8]: # Fill missing values

# Numeric columns -> median
df['retail_price'] = df['retail_price'].fillna(df['retail_price'].median())

df['discounted_price'] = df['discounted_price'].fillna(df['discounted_price'].median())

# Brand -> unknown
df['brand'] = df['brand'].fillna('unknown')

# Text/object columns -> unknown
text_columns = [
    'description',
    'image',
    'product_specifications',
    'product_name',
    'product_url',
    'product_category_tree',
    'pid'
]

for col in text_columns:
    df[col] = df[col].fillna('unknown')

# Ratings -> No Rating
df['product_rating'] = df['product_rating'].fillna('No Rating')

df['overall_rating'] = df['overall_rating'].fillna('No Rating')

# Boolean-like column
df['is_fk_advantage_product'] = (
    df['is_fk_advantage_product']
    .fillna(False)
)

# Timestamp
df['crawl_timestamp'] = (
    df['crawl_timestamp']
    .fillna('unknown')
)

# Check missing values again
print(df.isnull().sum())
```

uniq_id	1
crawl_timestamp	0
product_url	0
product_name	0
product_category_tree	0
pid	0
retail_price	0
discounted_price	0
image	0
is_fk_advantage_product	0
description	0
product_rating	0
overall_rating	0
brand	0
product_specifications	0
dtype: int64	

Screenshot: Missing Value Handling

Handling Missing Unique ID

Filled remaining missing unique ID values and rechecked missing counts.

```
# Fill missing uniq_id

df['uniq_id'] = df['uniq_id'].fillna('unknown_id')

# Verify
print(df.isnull().sum())
```

```
uniq_id                0
crawl_timestamp        0
product_url            0
product_name          0
product_category_tree  0
pid                   0
retail_price           0
discounted_price       0
image                 0
is_fk_advantage_product 0
description            0
product_rating         0
overall_rating         0
brand                 0
product_specifications 0
dtype: int64
```

Screenshot: Handling Missing Unique ID

Data Type Conversion

Converted timestamp and rating columns into useful analytical formats.

```
# Convert timestamp column

df['crawl_timestamp'] = pd.to_datetime(
    df['crawl_timestamp'],
    errors='coerce'
)

# Convert ratings to numeric
df['product_rating'] = pd.to_numeric(
    df['product_rating'],
    errors='coerce'
)

df['overall_rating'] = pd.to_numeric(
    df['overall_rating'],
    errors='coerce'
)

# Check updated data types
print(df.dtypes)
```

uniq_id	object
crawl_timestamp	datetime64[ns, UTC]
product_url	object
product_name	object
product_category_tree	object
pid	object
retail_price	float64
discounted_price	float64
image	object
is_fk_advantage_product	bool
description	object
product_rating	float64
overall_rating	float64
brand	object
product_specifications	object
dtype: object	

Screenshot: Data Type Conversion

Text Standardization

Standardized text columns using strip and lowercase operations.

```
# Standardize text columns

text_columns = df.select_dtypes(include='object').columns

for col in text_columns:
    df[col] = df[col].str.strip().str.lower()

# Check sample
df.head()
```

	uniq_id	crawl_timestamp	product_url	product_name	product_category_tree	pid	retail_price	dis
0	c2d766ca982eca8304150849735ffef9	2016-03-25 22:59:23+00:00	http://www.flipkart.com/alisha-solid-women-s-c...	alisha solid women's cycling shorts	["clothing >> women's clothing >> lingerie, sl...	srteh2ff9kedefgf	999.0	
1	7f7036a6d550aaa89d34c77bd39a5e48	2016-03-25 22:59:23+00:00	http://www.flipkart.com/fabhomedecor-fabric-do...	fabhomedecor fabric double sofa bed	["furniture >> living room furniture >> sofa b...	sbeeh3qgu7mfyjfy	32157.0	
2	f449ec65dcbc041b6ae5e6a32717d01b	2016-03-25 22:59:23+00:00	http://www.flipkart.com/aw-bellies/p/itmeh4grg...	aw bellies	["footwear >> women's footwear >> ballerinas >...	shoeh4grsubjgzxe	999.0	
3	0973b37acd0c664e3de26e97e5571454	2016-03-25 22:59:23+00:00	http://www.flipkart.com/alisha-solid-women-s-c...	alisha solid women's cycling shorts	["clothing >> women's clothing >> lingerie, sl...	srteh2f6huzmq6sj	699.0	
4	bc940ea42ee6bef5ac7cea3fb5cfbee7	2016-03-25 22:59:23+00:00	http://www.flipkart.com/sicons-all-purpose-arn...	sicons all purpose amica dog shampoo	["pet supplies >> grooming >> skin & coat care...	psoeh3zydmsyarj5	220.0	

Screenshot: Text Standardization

Rating Null Value Handling

Filled rating null values generated after numeric conversion.

```
# Fill rating NaN values with 0

df['product_rating'] = df['product_rating'].fillna(0)

df['overall_rating'] = df['overall_rating'].fillna(0)

# Verify
print(df[['product_rating', 'overall_rating']].isnull().sum())

product_rating    0
overall_rating    0
dtype: int64
```

Screenshot: Rating Null Value Handling

Discount Percentage Feature

Created discount_percentage using retail and discounted price.

```
# Create discount percentage column

df['discount_percentage'] = (
    (df['retail_price'] - df['discounted_price'])
    / df['retail_price']
) * 100

# Round values
df['discount_percentage'] = (
    df['discount_percentage']
    .round(2)
)

# Check sample
print(
    df[['retail_price',
        'discounted_price',
        'discount_percentage']].head()
)
```

	retail_price	discounted_price	discount_percentage
0	999.0	379.0	62.06
1	32157.0	22646.0	29.58
2	999.0	499.0	50.05
3	699.0	267.0	61.80
4	220.0	210.0	4.55

Screenshot: Discount Percentage Feature

Outlier Detection using IQR Method

Detected price outliers by calculating lower and upper bounds.

```
# Check outliers using IQR method

Q1 = df['retail_price'].quantile(0.25)
Q3 = df['retail_price'].quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - (1.5 * IQR)
upper_bound = Q3 + (1.5 * IQR)

outliers = df[
    (df['retail_price'] < lower_bound) |
    (df['retail_price'] > upper_bound)
]

print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)
print("Outliers Count:", outliers.shape[0])

Lower Bound: -1311.0
Upper Bound: 3985.0
Outliers Count: 2063
```

Screenshot: Outlier Detection using IQR Method

Price Outlier Flag Creation

Created a yes/no flag to identify price outlier products.

```
# Create outlier flag column

df['price_outlier'] = np.where(
    (df['retail_price'] > upper_bound) |
    (df['retail_price'] < lower_bound),
    'yes',
    'no'
)

# Check counts
print(df['price_outlier'].value_counts())
```

```
price_outlier
no      17938
yes      2063
Name: count, dtype: int64
```

Screenshot: Price Outlier Flag Creation

Final Dataset Validation

Verified final shape, missing values, data types, and sample data.

```
# Final dataset check

print("Final Shape:", df.shape)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nData Types:")
print(df.dtypes)

print("\nSample Data:")
print(df.head())

Final Shape: (20001, 17)

Missing Values:
uniq_id          0
crawl_timestamp  1
product_url      0
product_name     0
product_category_tree  0
pid             0
retail_price     0
discounted_price  0
image           0
is_fk_advantage_product  0
description      0
product_rating   0
overall_rating   0
brand           0
product_specifications  0
discount_percentage  0
price_outlier    0
dtype: int64

Data Types:
uniq_id          object
crawl_timestamp  datetime64[ns, UTC]
product_url      object
product_name     object
product_category_tree  object
pid             object
retail_price     float64
discounted_price float64
image           object
is_fk_advantage_product  bool
description      object
product_rating   float64
overall_rating   float64
brand           object
product_specifications  object
discount_percentage  float64
price_outlier    object
dtype: object

Sample Data:
   uniq_id  crawl_timestamp  \
0  c2d766c38928c38204150040725ff5f9  2015-02-25 22:59:22+00:00
```

Screenshot: Final Dataset Validation

Exporting Cleaned Dataset

Exported the final cleaned dataset into CSV format.

```
# Fill remaining timestamp null

df['crawl_timestamp'] = df['crawl_timestamp'].fillna(
    pd.Timestamp('2016-01-01', tz='UTC')
)

# Final check
print(df.isnull().sum())

# Export cleaned dataset
df.to_csv(
    "cleaned_flipkart_dataset.csv",
    index=False
)

print("Cleaned dataset exported successfully!")
```

```
uniq_id                0
crawl_timestamp         0
product_url            0
product_name           0
product_category_tree  0
pid                   0
retail_price           0
discounted_price       0
image                 0
is_fk_advantage_product 0
description            0
product_rating        0
overall_rating        0
brand                 0
product_specifications 0
discount_percentage    0
price_outlier         0
dtype: int64
Cleaned dataset exported successfully!
```

Screenshot: Exporting Cleaned Dataset

Loading Data into SQL Server

Loaded the cleaned dataset into SQL Server table flipkart_products.

```
# Convert timestamp column to string

df = df.rename(columns={'crawl_timestamp': 'crawl_datetime'})

table_name = "flipkart_products"

df.to_sql(
    "flipkart_products",
    engine,
    if_exists="replace",
    index=False
)

print(f"✓ Data successfully loaded into table '{table_name}'")

✓ Data successfully loaded into table 'flipkart_products'
```

Screenshot: Loading Data into SQL Server

Data Analysis using SQL Server

SQL Server was used to answer business questions after loading the cleaned dataset. The queries focused on product count, pricing, categories, brands, discount behavior, rating behavior, savings, and premium outlier products.

SQL analysis enabled query-based insights and supported Power BI dashboard design.

SQL Query Screenshots with Business Questions

1. What is the total number of products?

Counts total records available after cleaning.

The screenshot shows a SQL query in a text editor and its results in a table. The query is: `-- 1. Total Number of Products
SELECT COUNT(*) AS total_products
FROM flipkart_products;` The results table has one column, `total_products`, and one row with the value `20001`. The interface includes a zoom dropdown set to 100%, a status bar indicating 'No issues found', and tabs for 'Results' and 'Messages'.

	total_products
1	20001

Screenshot: 1. What is the total number of products?

2. What are the average retail price and average discounted price?

Compares original product price with selling price.

The screenshot shows a SQL query in a text editor and its results in a table. The query is: `-- 2. Average Retail Price and Discounted Price
SELECT
 AVG(retail_price) AS avg_retail_price,
 AVG(discounted_price) AS avg_discounted_price
FROM flipkart_products;` The results table has two columns, `avg_retail_price` and `avg_discounted_price`, and one row with the values `2971.54662266887` and `1967.77961101945`. The interface includes a zoom dropdown set to 100%, a status bar indicating 'No issues found', and tabs for 'Results' and 'Messages'.

	avg_retail_price	avg_discounted_price
1	2971.54662266887	1967.77961101945

Screenshot: 2. What are the average retail price and average discounted price?

3. Which categories have the highest number of products?

Identifies dominant product categories.

```
15
16  -- 3. Top Categories by Product Count
17  SELECT
18      product_category_tree,
19      COUNT(*) AS total_products
20  FROM flipkart_products
21  GROUP BY product_category_tree
22  ORDER BY total_products DESC;
23
```

100 % No issues found

Results Messages

	product_category_tree	total_products
1	["jewellery >> necklaces & chains >> necklaces"]	1567
2	["jewellery >> rings"]	710
3	["automotive >> accessories & spare parts >> car i..."]	522
4	["jewellery >> bangles, bracelets & armlets >> ban..."]	430
5	["tools & hardware >> tools >> gardening tools >> ..."]	332
6	["jewellery >> bangles, bracelets & armlets >> brac..."]	251
7	["mobiles & accessories >> tablet accessories >> c..."]	250
8	["clothing >> women's clothing >> lingerie, sleep &..."]	197
9	["home furnishing >> bed linen >> blankets, quilts ..."]	190
10	["footwear >> women's footwear >> wedges"]	186

Screenshot: 3. Which categories have the highest number of products?

4. Which brands have the highest number of products?

Finds top brands by product listing count.

```
24      -- 4. Top Brands by Number of Products
25      SELECT
26          brand,
27          COUNT(*) AS total_products
28      FROM flipkart_products
29      GROUP BY brand
30      ORDER BY total_products DESC;
31
32      -- 5. Average Discount Percentage by Category
```

100 %   No issues found

 Results  Messages

	brand	total_products
1	unknown	5865
2	allure auto	469
3	regular	314
4	voylla	299
5	slim	288
6	thelostpuppy	250
7	karatcraft	211
8	black	167
9	white	155
10	dailyobjects	144
11	speedwav	141
12	radiant bay	132
13	red	110
14	enthopia	101

Screenshot: 4. Which brands have the highest number of products?

5. Which categories have the highest average discount percentage?

Analyzes discount strategy by category.

```

32  -- 5. Average Discount Percentage by Category
33  SELECT
34      product_category_tree,
35      AVG(discount_percentage) AS avg_discount_percentage
36  FROM flipkart_products
37  GROUP BY product_category_tree
38  ORDER BY avg_discount_percentage DESC;
39

```

100 % No issues found

Results Messages

	product_category_tree	avg_discount_percentage
	["mobiles & accessories >> tablet accessories >> cas...	94.55
	["beauty and personal care >> makeup >> vanity boxe...	91.95
	["computers >> laptop accessories >> usb gadgets >...	91.72
	["mobiles & accessories >> mobile accessories >> bl...	91.6
	["baby care >> infant wear >> baby boys' clothes >> d...	90.9
	["clothing >> men's clothing >> winter & seasonal we...	90.57
	["computers >> laptop accessories >> usb gadgets >...	90.51666666666667
	["clothing >> women's clothing >> western wear >> s...	90.05
	["mobiles & accessories >> mobile accessories >> ot...	90.05
0	["gaming >> gaming accessories >> headphones (wit...	90.05
1	["bags, wallets & belts >> wallets & clutches >> travel ...	90.02
2	["jewellery >> accessories >> brooches >> remix yello...	88.43
3	["sunglasses >> barbarik wayfarer, aviator sunglasses"]	88.04
4	["mobiles & accessories >> mobile accessories >> he...	88.02
5	["clothing >> men's clothing >> accessories & combo ...	87.67
6	["watches >> wrist watches >> palito wrist watches >>...	86.73
7	["watches >> wrist watches >> palito wrist watches >>...	86.73

Screenshot: 5. Which categories have the highest average discount percentage?

6. Which products provide the highest discount percentage?

Finds extreme discount products.

```

40      -- 6. Products with Highest Discount Percentage
41      SELECT TOP 10
42          product_name,
43          brand,
44          retail_price,
45          discounted_price,
46          discount_percentage
47      FROM flipkart_products
48      ORDER BY discount_percentage DESC;
49

```

100 % ✓ No issues found

Results Messages

	product_name	brand	retail_price	discounted_price	discount_percentage
1	rajcrafts floral double quilts & comforters white	rajcrafts	6000	208	96.53
2	fashblush forever new orange bead & chain alloy n...	fashblush	3999	189	95.27
3	fashblush forever beautiful white bead alloy neckla...	fashblush	3499	189	94.6
4	bling book case for ipad 2 / 3	bling	1816	99	94.55
5	fashblush forever new blackbead & longchain allo...	fashblush	4999	279	94.42
6	fashblush alloy choker	fashblush	4499	289	93.58
7	fashblush forever new blackbead alloy necklace	fashblush	2999	199	93.36
8	fash blush forever new gold & white pearl chunk al...	fash blush	2999	199	93.36
9	mydress mystyle bulb led usb led light	mydress mystyle	999	75	92.49
10	fash blush forever new red & gold tribal thread allo...	fash blush	2499	198	92.08

Screenshot: 6. Which products provide the highest discount percentage?

7. Which are the most expensive products?

Identifies premium products and outliers.

```

49
50      -- 7. Most Expensive Products
51      SELECT TOP 10
52          product_name,
53          brand,
54          retail_price
55      FROM flipkart_products
56      ORDER BY retail_price DESC;
57
58      -- 8. Cheapest Products

```

100 % ✓ No issues found

Results Messages

	product_name	brand	retail_price
1	breitling ab011010/bb08 131s chronomat 44 analog ...	unknown	571230
2	durian leather 2 seater sofa	durian	250500
3	durian laze/3 leather 3 seater sofa	durian	217500
4	durian leather 2 seater sofa	durian	204600
5	cartier w6701005 analog watch - for boys, men	unknown	201000
6	durian club/3 leather 3 seater sofa	durian	162000
7	audeze lcd2 in bamboo, high quality planar magnetic...	audeze	116292
8	durian helena fabric 3 seater sofa	durian	108900
9	durian helena fabric 3 seater sofa	durian	108900
10	durian dream/2 leather 2 seater sofa	durian	108000

Screenshot: 7. Which are the most expensive products?

8. Which are the cheapest products?

Finds lowest priced products by discounted price.

```
58      -- 8. Cheapest Products
59      SELECT TOP 10
60          product_name,
61          brand,
62          discounted_price
63      FROM flipkart_products
64      ORDER BY discounted_price ASC;
65
```

100 % No issues found

Results Messages

	product_name	brand	discounted_price
1	geol wooden wet and dry broom	geol	35
2	havells havells - oro 32 one way electrical switch	havells	36
3	smartcraft generic netted art cloth pencil box	unknown	41
4	smartcraft generic netted art cloth pencil box	unknown	41
5	smartcraft generic netted art cloth pencil box	unknown	41
6	sam rolling pizza cutter	sam	45
7	khanna jupiter board & art paper box file	khanna jupiter	47
8	smartcraft angry birds cartoons art metal pencil ...	unknown	48
9	smartcraft generic pendant art cloth pencil box	unknown	48
10	maped twist and flex transparent rubber rulers	maped	49

Screenshot: 8. Which are the cheapest products?

9. What is the average rating by category?

Compares product quality perception across categories.

```
-- 9. Average Rating by Category
SELECT
    product_category_tree,
    AVG(product_rating) AS avg_rating
FROM flipkart_products
GROUP BY product_category_tree
ORDER BY avg_rating DESC;
```

100 %



No issues found



Results



Messages

	product_category_tree	avg_rating
1	bags, wallets & belts >> bags >> backpacks >> slb...	5
2	cameras & accessories >> camera accessories >...	5
3	beauty and personal care >> body and skin care >...	5
4	baby care >> baby bedding >> baby bedsheets >> ...	5
5	baby care >> baby bedding >> baby bedsheets >> ...	5
6	clothing >> kids' clothing >> girls wear >> dresses ...	5
7	baby care >> diapering & potty training >> nappies...	5
8	cameras & accessories >> camera accessories >...	5
9	clothing >> kids' clothing >> girls wear >> dresses ...	5
10	baby care >> baby bedding >> baby bedsheets >> ...	5

Screenshot: 9. What is the average rating by category?

10. Which brands have the highest average ratings?

Identifies high-rated brands.

```
73  
74 -- 10. Brands with Highest Average Rat  
75 SELECT  
76     brand,  
77     AVG(product_rating) AS avg_rating  
78 FROM flipkart_products  
79 GROUP BY brand  
80 ORDER BY avg_rating DESC;  
81
```

100 %

✓ No issues found

Results Messages

	brand	avg_rating
1	chipchop	5
2	alda	5
3	beadworks	5
4	balaji	5
5	jeet	5
6	crazytowear	5
7	bigshoponline	5
8	aarti collections	5
9	ladela	5
10	convenience	5
11	maharaja	5
12	aqua	5
13	lqrl	5

Screenshot: 10. Which brands have the highest average ratings?

11. Which products have high discount but low rating?

Finds products that may need quality review.

```

82  -- 11. Products with High Discount but Low Rating
83  SELECT
84      product_name,
85      brand,
86      retail_price,
87      discounted_price,
88      product_rating,
89      discount_percentage
90  FROM flipkart_products
91  WHERE discount_percentage > 50
92         AND product_rating < 3
93  ORDER BY product_rating ASC;

```

100 % No issues found

	product_name	brand	retail_price	discounted_price	product_rating	discount_percentage
1	alisha solid women's cycling shorts	alisha	999	379	0	62.06
2	aw bellies	aw	999	499	0	50.05
3	alisha solid women's cycling shorts	alisha	699	267	0	61.8
4	alisha solid women's cycling shorts	alisha	1199	479	0	60.05
5	dilli bazaaaar bellies, corporate casuals, casuals	dilli bazaaaar	699	349	0	50.07
6	alisha solid women's cycling shorts	alisha	1199	479	0	60.05
7	carrel printed women's	carrel	2299	910	0	60.42
8	alisha solid women's cycling shorts	alisha	999	379	0	62.06
9	alisha solid women's cycling shorts	alisha	999	379	0	62.06
10	style foot bellies	style foot	899	449	0	50.06

Screenshot: 11. Which products have high discount but low rating?

12. Which products have high rating but low discount?

Identifies premium products with strong ratings.

```

95  -- 12. Products with High Rating and Low Discount
96  SELECT
97      product_name,
98      brand,
99      retail_price,
100     discounted_price,
101     product_rating,
102     discount_percentage
103  FROM flipkart_products
104  WHERE product_rating >= 4
105         AND discount_percentage < 10
106  ORDER BY product_rating DESC;

```

100 % No issues found

	product_name	brand	retail_price	discounted_price	product_rating	discount_percentage
1	q&q lla2-213 digital watch - for boys, girls	unknown	645	645	5	0
2	carlton london boots	unknown	2295	2295	5	0
3	dearfoams boots	unknown	1820	1820	5	0
4	nishtaa yellow gold 22 k ring	nishtaa	7810	7420	5	4.99
5	betty girl's maxi dress	unknown	1995	1995	5	0
6	triumph form & beauty 30 w women's bra	deep lavender	899	899	5	0
7	posh women's layered dress	unknown	998	998	5	0
8	village handicrafts aluminum, ceramic necklace	village handicrafts	315	315	5	0
9	swethamber arts brass musical lady set of 4 statu...	swethamber arts	3500	3300	5	5.71
10	cookaid stainless steel with lid kadhai 1.2 l	cookaid	759	759	5	0

Screenshot: 12. Which products have high rating but low discount?

13. Which products give maximum savings?

Calculates savings using retail price minus discounted price.

```
107
108      -- 13. Products Giving Maximum Savings
109      SELECT TOP 10
110          product_name,
111          brand,
112          retail_price,
113          discounted_price,
114          (retail_price - discounted_price) AS savings
115      FROM flipkart_products
116      ORDER BY savings DESC;
117
```

100 %

✓ No issues found

Results Messages

	product_name	brand	retail_price	discounted_price	savings
1	durian leather 2 seater sofa	durian	250500	162825	87675
2	durian laze/3 leather 3 seater sofa	durian	217500	141375	76125
3	durian leather 2 seater sofa	durian	204600	132990	71610
4	durian club/3 leather 3 seater sofa	durian	162000	105300	56700
5	durian helena fabric 3 seater sofa	durian	108900	70785	38115
6	durian helena fabric 3 seater sofa	durian	108900	70785	38115
7	durian dream/2 leather 2 seater sofa	durian	108000	70200	37800
8	durian fabric 3 seater sofa	durian	93600	60840	32760
9	durian harmony/2 fabric 2 seater sofa	durian	85500	55575	29925
10	durian fabric 2 seater sofa	durian	84300	54795	29505

Screenshot: 13. Which products give maximum savings?

14. How many products are available in each category?

Shows product distribution by category.

```

117
118 -- 14. Number of Products in Each Cat
119 SELECT
120     product_category_tree,
121     COUNT(*) AS total_products
122 FROM flipkart_products
123 GROUP BY product_category_tree
124 ORDER BY total_products DESC;
125

```

100 % No issues found

Results Messages

	product_category_tree	total_products
1	["jewellery >> necklaces & chains >> necklaces"]	1567
2	["jewellery >> rings"]	710
3	["automotive >> accessories & spare parts >> car i...	522
4	["jewellery >> bangles, bracelets & armlets >> ban...	430
5	["tools & hardware >> tools >> gardening tools >> ...	332
6	["jewellery >> bangles, bracelets & armlets >> brac...	251
7	["mobiles & accessories >> tablet accessories >> c...	250
8	["clothing >> women's clothing >> lingerie, sleep &...	197
9	["home furnishing >> bed linen >> blankets, quilts ...	190
10	["footwear >> women's footwear >> wedges"]	186
11	["jewellery >> jewellery sets"]	161
12	["mobiles & accessories >> tablet accessories >> c...	144
13	["clothing >> men's clothing >> t-shirts >> numero ...	135
14	["home furnishing >> curtains & accessories >> cur...	131
15	["footwear >> women's footwear >> casual shoes >...	118
16	["clothing >> women's clothing >> lingerie, sleep &...	101
17	["mobiles & accessories >> tablet accessories >> c...	101
18	["footwear >> women's footwear >> heels"]	97
19	["kitchen & dining >> coffee mugs >> prithish coffe...	94
20	["jewellery >> earrings"]	93
21	["home furnishing >> bath linen >> towels"]	92

Screenshot: 14. How many products are available in each category?

15. Which category has the highest average selling price?

Compares average discounted price by category.

126	-- 15. Category with Highest Average Selling Pri
127	SELECT
128	product_category_tree,
129	AVG(discounted_price) AS avg_selling_price
130	FROM flipkart_products
131	GROUP BY product_category_tree
132	ORDER BY avg_selling_price DESC;
133	

100 %
No issues found

Results
Messages

	product_category_tree	avg_selling_price
1	["watches >> wrist watches >> breitling wrist watches"]	571230
2	["watches >> wrist watches >> cartier wrist watches"]	201000
3	["mobiles & accessories >> mobile accessories >> h..."]	116292
4	["watches >> wrist watches >> tag heuer wrist watch..."]	107750
5	["clothing >> men's clothing >> winter & seasonal w..."]	48999
6	["pens & stationery >> office equipments >> biometri..."]	44804
7	["furniture >> living room >> sofas & sectionals"]	42137.6610169492
8	["computers >> laptops >> hp laptops"]	37140
9	["pens & stationery >> office equipments >> biometri..."]	36575
10	["computers >> network components >> routers >> u..."]	31000
11	["furniture >> bedroom furniture >> mattresses >> sp..."]	26530.5
12	["furniture >> living room furniture >> tv & entertainm..."]	25380
13	["furniture >> bedroom furniture >> beds >> ethnic h..."]	24999
14	["furniture >> bedroom furniture >> beds >> ethnic h..."]	24999
15	["clothing >> women's clothing >> ethnic wear >> le..."]	24995

Screenshot: 15. Which category has the highest average selling price?

16. Which products are above category average price?

Finds relatively expensive products within their category.

```

134  -- 16. Products Above Category Average Price
135  SELECT
136      product_name,
137      brand,
138      discounted_price,
139      product_category_tree
140  FROM flipkart_products p
141  WHERE discounted_price >
142      (
143          SELECT AVG(discounted_price)
144          FROM flipkart_products
145          WHERE product_category_tree = p.product_category_tree
146      );

```

JYOTHSNA (SQL Server 17.0.1000.7 - JYOTHSNA\jyoth)

100 % No issues found

Results Messages

	product_name	brand	discounted_price	product_category_tree
1	salt n pepper 12-298 taupe boots	unknown	2150	["footwear >> women's footwear >> casual shoes ...
2	magnum footwear lifestyle boots	unknown	3199	["footwear >> women's footwear >> casual shoes ...
3	steppings trendy boots	unknown	2519	["footwear >> women's footwear >> casual shoes ...
4	tsg kelz boots	unknown	2650	["footwear >> women's footwear >> casual shoes ...
5	alisha solid women's cycling shorts	alisha	479	["clothing >> women's clothing >> lingerie, sleep &...
6	get glamr designer snowra boots	unknown	2239	["footwear >> women's footwear >> casual shoes ...
7	lenco dcompass2010 quazi analog-digital watch - f...	unknown	849	["watches >> wrist watches >> lenco wrist watches"]
8	alisha solid women's cycling shorts	alisha	479	["clothing >> women's clothing >> lingerie, sleep &...
9	pinza black christmas boots	unknown	1900	["footwear >> women's footwear >> casual shoes ...
10	karatcraft duilla ruby gold diamond 18 k ring	karatcraft	42262	["jewellery >> rings"]
11	karatcraft journa gold diamond 18 k ring	karatcraft	37787	["jewellery >> rings"]
12	salt n pepper 13-552 rebacca almond boots	unknown	2621	["footwear >> women's footwear >> casual shoes ...
13	caratlane fantasia band white gold diamond 18 k ring	caratlane	31033	["jewellery >> rings"]
14	vastra valley girl's a-line dress	unknown	797	["clothing >> kids' clothing >> girls wear >> dresse...
15	disney dw100405 digital watch - for boys	unknown	1345	["watches >> wrist watches >> disney wrist watches"]

Screenshot: 16. Which products are above category average price?

17. Which products are below category average price?

Finds relatively cheaper products within their category.

```

147  -- 17. Products Below Category Average Price
148  SELECT
149      product_name,
150      brand,
151      discounted_price,
152      product_category_tree
153  FROM flipkart_products p
154  WHERE discounted_price <
155      (
156          SELECT AVG(discounted_price)
157          FROM flipkart_products
158          WHERE product_category_tree = p.product_category_tree
159      );
160

```

100 % No issues found

Results Messages

	product_name	brand	discounted_price	product_category_tree
1	steppings trendy boots	unknown	1099	["footwear >> women's footwear >> casual shoes ..."]
2	j.d.h. 111-222 boots	unknown	599	["footwear >> women's footwear >> casual shoes ..."]
3	disney 98189 analog watch - for boys, girls	unknown	900	["watches >> wrist watches >> disney wrist watches"]
4	herberto girl's high low dress	unknown	1025	["clothing >> kids' clothing >> girls wear >> dresse..."]
5	herberto girl's a-line dress	unknown	949	["clothing >> kids' clothing >> girls wear >> dresse..."]
6	kielz ladies boots	unknown	1299	["footwear >> women's footwear >> casual shoes ..."]
7	selfie boots	unknown	599	["footwear >> women's footwear >> casual shoes ..."]
8	steppings trendy boots	unknown	719	["footwear >> women's footwear >> casual shoes ..."]
9	faballey casual short sleeve solid women's top	unknown	750	["clothing >> women's clothing >> western wear >..."]
10	belle gambe boots	unknown	1795	["footwear >> women's footwear >> casual shoes ..."]
11	sdfs casual, formal, party sleeveless self design w...	unknown	849	["clothing >> women's clothing >> western wear >..."]
12	kennel rubber dumbell with bell - small rubber ru...	kennel	190	["pet supplies >> toys >> comfort toys"]
13	disney lp-1008 (yellow) analog watch - for boys, g...	unknown	795	["watches >> wrist watches >> disney wrist watches"]
14	willy winkies black boots	unknown	999	["footwear >> women's footwear >> casual shoes ..."]
15	faballey casual sleeveless solid women's top	unknown	900	["clothing >> women's clothing >> western wear >..."]
16	djh boots	unknown	499	["footwear >> women's footwear >> casual shoes ..."]
17	faballey casual sleeveless solid women's top	unknown	900	["clothing >> women's clothing >> western wear >..."]
18	alisha solid women's cycling shorts	alisha	379	["clothing >> women's clothing >> lingerie, sleep &..."]
19	allen solly girl's gathered dress	unknown	1119	["clothing >> kids' clothing >> girls wear >> dresse..."]
20	mvnte solid women's cvclina shorts. avm shorts. s...	mvnte	649	["clothing >> women's clothing >> sports & avm we..."]

Screenshot: 17. Which products are below category average price?

18. What is the rating distribution of products?

Shows count of products for each rating level.

```
160
161      -- 18. Rating Distribution
162      SELECT
163          product_rating,
164          COUNT(*) AS total_products
165      FROM flipkart_products
166      GROUP BY product_rating
167      ORDER BY product_rating DESC;
168
169      -- 19. Brands with High Ratings and High Discounts
170      SELECT
171          brand,
172          AVG(product_rating) AS avg_rating
```

100 % No issues found

Results Messages

	product_rating	total_products
1	5	620
2	4.9	4
3	4.8	21
4	4.7	24
5	4.6	9
6	4.5	67
7	4.4	16
8	4.3	45
9	4.2	47
10	4.1	24
11	4	246
12	3.9	15
13	3.8	23
14	3.7	51
15	3.6	25
16	3.5	45
17	3.4	13
18	3.3	17

Screenshot: 18. What is the rating distribution of products?

19. Which brands have both high ratings and high discounts?

Finds strong value brands based on rating and discount.

```
168
169 -- 19. Brands with High Ratings and High Discounts
170 SELECT
171     brand,
172     AVG(product_rating) AS avg_rating,
173     AVG(discount_percentage) AS avg_discount
174 FROM flipkart_products
175 GROUP BY brand
176 HAVING AVG(product_rating) >= 4
177 ORDER BY avg_discount DESC;
```

100 % No issues found

Results Messages

	brand	avg_rating	avg_discount
1	bond beatz	5	91.6
2	kaizer jewelry	5	89.62
3	bracketron	4	87.94
4	sharani	5	80.1
5	raees	5	79.23
6	dreamshop	5	78.46
7	nihar	5	75.72
8	sb retails	5	75.08
9	riana	5	74.03
10	stybuzz	5	73.45
11	ambitione	5	70.78
12	dark blue	4	70.1
13	pure source	5	70
14	turquoise	5	67.97
15	jewels guru	5	67.67

Screenshot: 19. Which brands have both high ratings and high discounts?

20. Which are the best value products?

Combines rating and discount to identify best products.

```

179  -- 20. Best Value Products
180  SELECT TOP 20
181      product_name,
182      brand,
183      product_rating,
184      retail_price,
185      discounted_price,
186      discount_percentage
187  FROM flipkart_products
188  WHERE product_rating >= 4
189  ORDER BY discount_percentage DESC, product_rating DESC;

```

100 % No issues found

Results Messages

	product_name	brand	product_rating	retail_price	discounted_price	discount_percentage
1	bond beatz zipper with excellent sound wired head...	bond beatz	5	2499	210	91.6
2	elligator 10 card holder	unknown	4	2995	299	90.02
3	kaizer jewelry stainless steel zircon ring	kaizer jewelry	5	5769	599	89.62
4	bracketron tour window hip kicker	bracketron	4	1650	199	87.94
5	jack klein goldjack analog watch - for boys, men	unknown	5	1999	262	86.89
6	younky women's animal print casual shirt	unknown	5	1999	349	82.54
7	rodid full sleeve solid men's sweatshirt	unknown	4.3	2080	399	80.82
8	jewellerywale alloy necklace	jewellerywale	5	2300	449	80.48
9	etti esj69_multi glass necklace	etti	5	1000	199	80.1
10	sharani yellow gold yellow gold 22 bangle	sharani	5	1000	199	80.1
11	karishma women's solid formal shirt	unknown	5	999	199	80.08
12	younky women's sports bra	black	4	999	199	80.08
13	ethnic jewels alloy yellow gold bangle set	ethnic jewels	5	1619	323	80.05
14	top notch solid women's polo neck t-shirt	unknown	5	1499	299	80.05

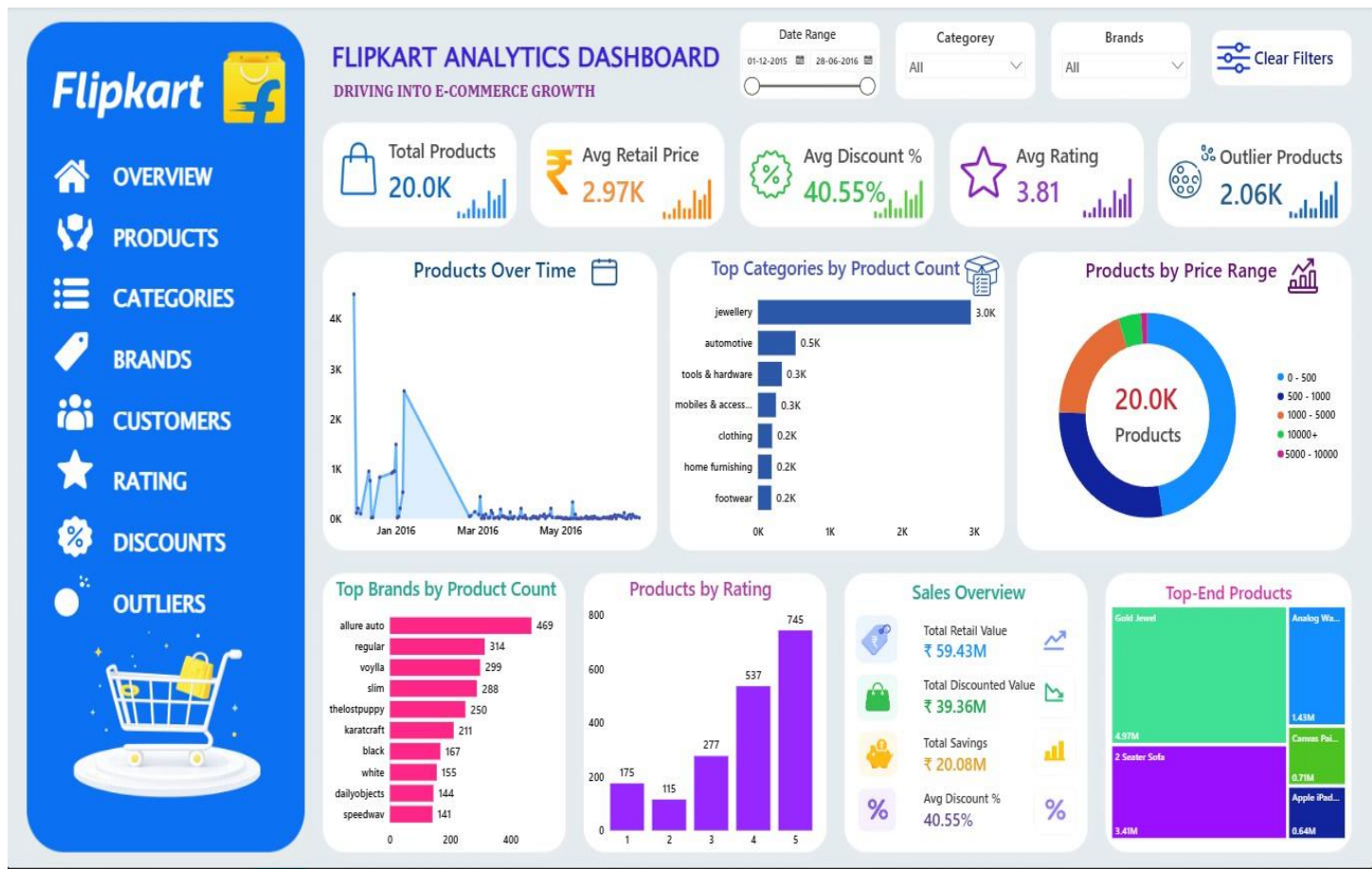
Screenshot: 20. Which are the best value products?

Power BI Dashboard

The final dashboard was created in Power BI. It contains KPI cards, filters, product trends, top categories, price range distribution, top brands, rating distribution, sales overview, and top-end product treemap.

Dashboard Components

- **KPI Cards:** Total Products, Average Retail Price, Average Discount %, Average Rating, Outlier Products.
- **Filters:** Date Range, Category, Brand, Clear Filters button.
- **Charts:** Products Over Time, Top Categories, Products by Price Range, Top Brands, Products by Rating.
- **Sales Overview:** Total Retail Value, Total Discounted Value, Total Savings, Average Discount, Outlier Products.
- **Treemap:** Top-end products by retail price.



Final Power BI Dashboard Screenshot

Business Insights

- The cleaned dataset contains 20,001 products and 17 columns.
- Average retail price is approximately INR 2.97K, while the average discounted price is approximately INR 1.97K.
- Average discount is 40.55%, showing strong discounting across many products.
- Average overall rating becomes meaningful after excluding unrated products, resulting in 3.81.
- 2,063 products were identified as price outliers using the IQR method.
- Luxury watches, premium sofas, paintings, jewellery, and Apple iPad Air products appeared as top-end product groups.
- Top categories and top brands help understand product distribution across Flipkart listings.
- Sales overview helps compare potential retail value, actual discounted value, and customer savings.

Conclusion

The Flipkart Product Analytics Dashboard project successfully demonstrates an end-to-end data analytics workflow. Python was used for data cleaning and feature engineering, SQL Server was used for query-based business analysis, and Power BI was used to build a modern interactive dashboard. The project provides useful insights into pricing, discounts, product ratings, categories, brands, and top-end product outliers.

Future Scope

- Add customer behavior and sales transaction data to perform revenue forecasting.
- Use recommendation models to suggest best-value products.
- Add product image analytics using image URLs.
- Integrate live dashboard refresh from SQL Server.
- Create advanced drill-through reports for category and brand analysis.

References

- Kaggle Flipkart e-commerce product dataset.
 - Python Pandas documentation for data cleaning and preprocessing.
 - Microsoft SQL Server documentation for query-based analysis.
 - Microsoft Power BI documentation for data visualization and dashboard creation.
-

END REPORT
